



Solving Polynomial Equations

To solve a polynomial equation, set the equation equal to zero and factor completely. Then apply the zero product property to find the solutions of the equation.

Zero Product Property

The zero product property states that if the product of two or more expressions is zero, then one or more of the expressions are equal to zero.

Solving a Polynomial Equation in Factored Form

Set each factor equal to zero to find all the solutions of the polynomial equation.

EXAMPLE 1

Solve the equation.

A $(x + 1)(5x - 4) = 0$

SOLUTION

Set each factor equal to zero and solve.

$$\begin{array}{ll} x + 1 = 0 & 5x - 4 = 0 \\ x = -1 & 5x = 4 \\ & x = \frac{4}{5} \end{array}$$

The solution set is $\left\{-1, \frac{4}{5}\right\}$.

► **Think About It** Always be certain that the original equation is set equal to zero before solving.

B $3(2x + 1)(x - 2) = 0$

SOLUTION

For each factor containing a variable, set the factor equal to zero and solve.

$$\begin{array}{rcl} 2x + 1 = 0 & & x - 2 = 0 \\ 2x = -1 & & x = 2 \\ x = -\frac{1}{2} & & \end{array}$$

The solution set is $\left\{-\frac{1}{2}, 2\right\}$. ■

Using Factoring Patterns to Solve a Polynomial Equation

You can use factoring patterns to help solve some polynomial equations, but do not try to factor and use the zero product property unless the equation has zero alone on one side.

EXAMPLE 2

Use factoring and the zero product property to solve the equation.

A $4x^2 - 16 = 9$

SOLUTION

Set the equation equal to zero first and then factor the polynomial.

$$4x^2 - 16 = 9$$

$$4x^2 - 25 = 0 \quad \text{Transform the equation so it has zero alone on one side of the equals sign.}$$

$$(2x - 5)(2x + 5) = 0 \quad \text{Factor the difference of squares.}$$

Apply the zero product property.

$$2x - 5 = 0$$

$$2x = 5$$

$$x = \frac{5}{2}$$

$$2x + 5 = 0$$

$$2x = -5$$

$$x = -\frac{5}{2}$$

The solution set is $\left\{-\frac{5}{2}, \frac{5}{2}\right\}$.

B $2x^3 + 4x^2 + 2x = 0$

SOLUTION

Factor the polynomial completely and then solve the equation.

$$2x^3 + 4x^2 + 2x = 0$$

$$2x(x^2 + 2x + 1) = 0 \quad \text{Factor out the greatest common monomial.}$$

$$2x(x + 1)(x + 1) = 0 \quad \text{Factor the perfect square trinomial.}$$

Apply the zero product property.

$$2x = 0$$

$$x = 0$$

$$x + 1 = 0$$

$$x = -1$$

$$x + 1 = 0$$

$$x = -1$$

The solution set is $\{-1, 0\}$.

► **Think About It** Because the factor $(x + 1)$ appears two times in the factored form of the equation, -1 is a solution with multiplicity 2.

C $x^3 + 3x^2 - 4x - 12 = 0$

SOLUTION

Factor by grouping and then solve the equation.

$$x^3 + 3x^2 - 4x - 12 = 0$$

$$(x^3 + 3x^2) + (-4x - 12) = 0 \quad \text{Group monomials with a common factor.}$$

$$x^2(x + 3) + (-4)(x + 3) = 0 \quad \text{Factor out the greatest common monomial for each group.}$$

$$(x^2 - 4)(x + 3) = 0 \quad \text{Distributive Property}$$

$$(x + 2)(x - 2)(x + 3) = 0 \quad \text{Factor the difference of squares.}$$

Apply the zero product property.

$$x + 2 = 0$$

$$x = -2$$

$$x - 2 = 0$$

$$x = 2$$

$$x + 3 = 0$$

$$x = -3$$

The solution set is $\{-3, -2, 2\}$.

D $8x^4 - 15x^3 = 2x^2$

► **Remember** To solve a polynomial equation, set the equation equal to zero first and then factor the polynomial.

SOLUTION

Factor out any common factors. Factor the remaining trinomial. Then solve the equation.

$$8x^4 - 15x^3 - 2x^2 = 0$$

$$x^2(8x^2 - 15x - 2) = 0 \quad \text{Factor out the GCF.}$$

$$x^2(8x + 1)(x - 2) = 0 \quad \text{Distributive Property}$$

Apply the zero product property.

$$x^2 = 0$$

$$x = 0$$

$$8x + 1 = 0$$

$$8x = -1$$

$$x = -\frac{1}{8}$$

$$x - 2 = 0$$

$$x = 2$$

The solution set is $\left\{-\frac{1}{8}, 0, 2\right\}$. ■

Checking the Solutions of a Polynomial Equation

To verify that the solutions found are correct, substitute each one in the original equation. All correct solutions will make the original equation true.

EXAMPLE 3

Use factoring to solve the equation and check the solutions.

$$(2x + 1)^3 - (2x + 1) = 0$$

SOLUTION

Factor the polynomial completely.

$$(2x + 1)^3 - (2x + 1) = 0$$

$$(2x + 1) \left[(2x + 1)^2 - 1 \right] = 0 \quad \text{Factor out the common binomial.}$$

$$(2x + 1) \left[(2x + 1) - 1 \right] \left[(2x + 1) + 1 \right] = 0 \quad \text{Factor the difference of squares inside the brackets.}$$

$$(2x + 1)(2x)(2x + 2) = 0 \quad \text{Simplify inside the brackets.}$$

Apply the zero product property.

$$2x + 1 = 0$$

$$2x = 0$$

$$2x + 2 = 0$$

$$2x = -1$$

$$x = 0$$

$$2x = -2$$

$$x = -\frac{1}{2}$$

$$x = -1$$

Check each solution by substituting into the original equation.

$$\text{Let } x = -\frac{1}{2}.$$

$$\begin{aligned} (2x + 1)^3 - (2x + 1) &= \left(2 \cdot \left(-\frac{1}{2} \right) + 1 \right)^3 - \left(2 \cdot \left(-\frac{1}{2} \right) + 1 \right) \\ &= (-1 + 1)^3 - (-1 + 1) \\ &= 0 - 0 \\ &= 0 \checkmark \end{aligned}$$

Let $x = 0$.

$$\begin{aligned}(2x + 1)^3 - (2x + 1) &= (2 \bullet 0 + 1)^3 - (2 \bullet 0 + 1) \\ &= 1^3 - 1 \\ &= 1 - 1 \\ &= 0 \checkmark\end{aligned}$$

Let $x = -1$.

$$\begin{aligned}(2x + 1)^3 - (2x + 1) &= (2 \bullet (-1) + 1)^3 - (2 \bullet (-1) + 1) \\ &= (-2 + 1)^3 - (-2 + 1) \\ &= (-1)^3 - (-1) \\ &= -1 + 1 \\ &= 0 \checkmark\end{aligned}$$

Each solution checks out correctly. The solution set is $\left\{-1, -\frac{1}{2}, 0\right\}$. ■